CS 3853 Computer Architecture

Project Report

Sun, Nov 1st, 2026 11:59 p.m.

Team 4

Group Members

Com Morales, Paula Deyanira
Espinoza, Francisco
Meier, Mercy Jubilee
Suarez Munoz, Ana Julietee

By signing this report I affirm that I know and agree with the contents.

Signatures:

Name #1: Ana Suarez Munoz

Name #2: Mercy Meier

Name #3: Paula Com Morales

Name #4: Francisco Espinoza

Brief Overview:
Our objective for Milestone #1 was to develop the initial foundation of a virtual memory and cache simulator for a 32-bit CPU. Our team implemented a C program that would accept command-line input parameters and calculate the corresponding cache and physical memory characteristics.

The program accepts cache size, block size, associativity, replacement policy, physical memory size, operating system memory usage, instructions per time slice, and trace file names. It then calculates the total cache blocks, tag and index sizes, number of rows, overhead, implementation memory size, cost, physical pages, and page table memory requirements.

We generated three simulation outputs using different configurations to demonstrate how the calculated values change depending on the input parameters. The milestone provides the foundation for implementing virtual memory operations and cache simulation in the following milestones.

Technical Issues:

One technical challenge involved calculating the cache tag and index sizes correctly for different configurations. These values depend on the cache size, block size, associativity, and physical memory address width. Our implementation uses logarithmic calculations to determine the number of index and offset bits and subtracts those values from the physical address width to calculate the tag size.

Another important consideration was calculating the total memory required for page tables. Since the simulator supports multiple trace files, the calculation must account for the number of processes represented by the supplied files.

Group Member Contributions:

A. **Com Morales, Paula Deyanira:** Parsed the arguments and calculated the cache values.
B. **Espinoza, Francisco:** Calculated the physical values.
C. **Meier, Mercy Jubilee:** Gathered the textual output of the three different runs.
D. **Suarez Munoz, Ana Juliette:** Drafted the report and created the executable file.

Group Issues and Resolutions:

None.